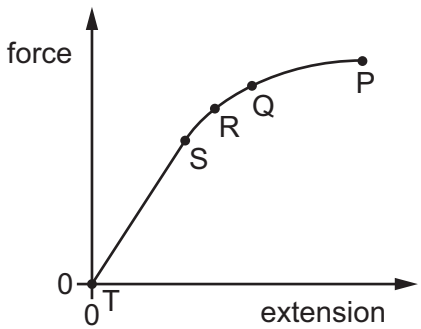
**21** The extension of a copper wire is measured for different forces applied to the wire. A graph is plotted to show the variation of the force on the wire against extension. The maximum force is applied at point P.




Which statement **must** be correct?

**A** Point R is the limit of proportionality.

**B** The elastic potential energy of the wire at point S is given by the area under the graph between points T and S.

**C** There is no plastic deformation between points Q and P.

**D** The wire obeys Hooke's law up to a point between R and Q.